A UAV-Aided Digital Twin Framework for IoT Networks With High Accuracy and Synchronization

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***Abstract*—Digital Twin (DT) technology has emerged as a promising link between the physical and virtual worlds, enabling simulation, prediction, and real-time performance optimization in different domains. In this work we develop a high-fidelity digital twin framework, focusing on synchronization and accuracy between physical and digital systems to enhance data-driven decision making. To achieve this, we deploy several stationary UAVs in optimized locations to collect data from IoT devices, which were used to monitor multiple physical entities and perform computations to evaluate their status. We formulate a mixed-integer non-convex program to maximize the total amount of data collected from all IoT devices while ensuring a constrained age of digital twin threshold and solve it using successive convex approximation (SCA). To cope with realistic scenarios involving unpredictable environments and large network sizes, we model our problem as a Markov Decision Process (MDP), and propose a deep reinforcement learning-based approach using a Twin Delayed Deep Deterministic Policy Gradient (TD3) to optimize the uncrewed aerial vehicle positions and the sum rate. Finally, we present different simulation results of the SCA and TD3 based solutions together with two baseline approaches and evaluated the sum rate in terms of IoT device count, AoDT threshold, task arrival rate and UAVs' computational capacity. In all simulation results, the proposed TD3-based approach consistently proved to be superior as compared to the baseline solutions.**

***Index Terms*—Age of information, digital twin, high accuracy and synchronization, age of digital twin, uncrewed aerial vehicles, multiple-sensor monitoring.**

I. Introduction

With the emerging 5G networks that ensure instantaneous connectivity to billions of Internet of Things devices, the demand for accurate and fast data updates is increasing to facilitate remote monitoring and control [1], [2]. This presents significant challenges for network operators in providing dynamic adaptation to customer needs. To overcome these challenges, many industries are using digital twins to improve operational efficiency and build accurate maintenance strategies. A recent study estimates that 85% of IoT industries are expected to incorporate digital twinning in the near future [3].

A digital twin is a virtual representation of a physical system, process, or object designed to accurately reflect the physical entity through synchronization and real-time data updates. It uses simulation, machine learning, and reasoning to make better decisions [4], [5]. Digital Twin platforms provide networks with improved automation, resilience testing, full life cycle operation, and infrastructure maintenance [5]. To build a high-fidelity digital twin, various challenges are involved. According to [5] and [6], the main challenges include data acquisition and processing, high-fidelity modeling, and real-time communication between virtual and real twins. The precision and efficiency of data collection significantly affect the quality of the DT model [7]. Since a large amount of real-time data is required to establish a high-fidelity digital twin [5], IoT devices often fail to deliver their real-time measurements to the base station due to limited transmit power and processing capacity. Utilizing uncrewed aerial vehicles (UAVs) to collect and process data from IoT devices offers a reliable and affordable wireless connectivity solution and is widely adopted in similar problem setup [8], [9], [10], [11].

The large amount of real-time data enforces adequate synchronization and accuracy between the physical and digital twins. On one hand, synchronization can be improved by reducing the Age of Digital Twin (AoDT) metric. This synchronization metric measures the time gap between the current status of the monitored physical entity and its digital counterpart. AoDT was first introduced in [12] and is based on the AoI, which reflects information freshness and was first introduced in 2010 [13]. Thus, AoDT measures status freshness to determine whether the digital twin accurately mirrors the current condition of a given physical entity. On the other hand, a higher level of accuracy can be achieved by maximizing the amount of data collected. However, jointly enhancing synchronization and accuracy introduces a tradeoff since we are considering several IoT devices monitoring different physical processes. While attempting to improve the accuracy of one physical entity by collecting more data, the synchronization of another physical entity is negatively impacted, resulting in a tradeoff between the accuracy and synchronization among the digital twins of the various physical entities. Thus, higher accuracy increases AoDT, leading to a lack of synchronization between the digital and physical twins. To this end, this work develops a digital twin framework that

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optimizes a DT-aided IoT network by balancing accuracy and synchronization.

We consider a system composed of several physical entities, each monitored by a group of IoT devices. UAVs are deployed to collect measurements from all IoT devices and process them to determine the status of the physical entities in the system. Our work utilizes the novel AoDT metric, which accounts for both upload and processing of sensitive data through employment of UAVs as edge devices to provide computing services and efficient data collection and transmission to the base station. We derive a closed-form expression for the AoDT metric to reflect the status freshness of the digital twin. We then formulate a mixed-integer non-convex optimization problem to maximize DT accuracy while maintaining synchronization and solve the problem using successive convex approximations (SCA). We propose a more efficient and scalable solution using reinforcement learning, specifically twin delayed deep deterministic policy gradient (TD3) solution approach. Extensive simulations validate the effectiveness of the SCA-based and TD3-based approaches. To the best of our knowledge, no research work has tackled a similar problem with IoT devices sending fresh status updates to the digital twin platform to build a highly synchronized and accurate digital twin, with each physical entity being monitored by multiple IoT devices.

The paper is organized as follows. We start by surveying recent relevant literature in Section II and highlight our contributions. Section III presents the system model and its key components. Section IV formulates the problem as a mixed integer non-convex program, and Section V proposes the convex optimization solution approach while Section VI proposes the TD3 model solution approach. Simulation results and analysis are provided in Section VII. Finally, Section VIII concludes the work and presents interesting future directions.

II. Literature Review

To build a high-fidelity digital twin, various challenges are involved. According to [5] and [6], the main challenges include data acquisition and processing, high-fidelity modeling, and real-time communication between virtual and real twins. The precision and efficiency of data collection significantly affect the quality of the DT model [7]. In this section, we survey selected related literature that utilizes mobile edge computing in delay sensitive IoT applications to empower the DT technology. Authors in [12] developed a high-fidelity digital twin model where its consistency with the physical system is measured through accuracy and synchronization. They deployed a UAV that follows an optimized trajectory to collect data from multiple industrial IoT devices. The data was then processed through several virtual machines equipped in the UAV to determine the status of physical entities, which was subsequently uploaded to the digital twin running at the base station. Authors introduced the age of digital twin (AoDT) metric to represent the synchronization gap between the actual status of a physical entity and its digital replica. The main aim was to develop a DT model consistent with the physical system, relying on the amount of data collected and on AoDT as crucial metrics to enhance DT accuracy and synchronization. Authors only utilized one UAV and did not consider any practical scenario of equipping devices with multiple sensors. Moreover authors did not address scalability or complex industrial IoT environments. In [14], authors aim to optimize the trajectories of multiple UAVs serving several IIoT devices in an industrial urban field using DT technology. The main aim of the study is to minimize the total energy consumption and improve the efficiency of UAVs in terms of latency and computation. Several IIoTs devices with different tasks; classified as delay sensitive and computational sensitive tasks, are deployed in different subareas. Each IIoT device sends its requirement state data via the UAV to the DT server, which updates the global state of the system and assigns UAVs to perform these tasks. The authors proposed dueling deep Q-networks with prioritized experience replay to optimize trajectory planning and balance latency, computation, and energy. Li et al. in [15] studied the problem of synchronizing the digital twin with its physical counterpart using an energy-constrained UAV that collects data from sensors deployed in a remote region. Each sensor is powered by a solar panel and its generated data is energy constrained. Authors assume that the digital twin is updated in discretized time slots. Given that the DTs of sensors are stored in cloudlets in a mobile edge computing network, authors introduced an update budget that reflects how much DT states of a specific group of sensors are synchronized with their physical counterparts at each time slot. The aim is to minimize the average DT state staleness of all objects given an energy-limited UAV. However in both problems ([14] and [15]), authors assumed that each device is equipped with one sensor, age of information metric was not utilized for freshness and authors only considered one traveling UAV. Moreover, in [15], authors considered partial DT object synchronization per round instead of maximizing data collected from all devices for optimum accuracy with synchronization. Authors in [16] studied the problem of digital twin placement for improving user satisfaction based on minimum user query service delays that depends on digital twin data freshness which, in turn, relies on the age of information. They proposed two different problem scenarios: static digital twin placement and dynamic digital twin placement based on launching new digital twins periodically. Both scenarios involved deploying digital twins in cloudlets to maximize user satisfaction based on information freshness and subject to computational capacity constraints. In [17], authors proposed a digital twin edge network where two types of real-time tasks in digital twin applications are considered: update tasks for periodic synchronization with the digital twin, and inference tasks for testing the response of a physical system by simulating the corresponding digital twin by users. The system consisted of an edge server with a set of physical systems, and multiple users. For each physical system, the corresponding cyber twin is implemented in the edge server to provide DT applications. Update tasks are generated by data collected from the physical systems while inference tasks are requested by users on demand. Authors formulated an optimization problem to maximize digital twin freshness ratio. Authors proposed a DT task scheduler that prioritizes update tasks based on the number of corresponding inference tasks. Both studies ([16], [17]) utilized age of information metric for DT freshness and

TABLE I
RELATED WORK COMPARISON

Reference Number	AoI Metric	DT	UAV(s) Deployment	UAV(s) Motion	Algorithm Used	Objective
[11]	✓		Single	✓	Successive convex approximation	AoI-Based UAV trajectory planning given energy constraint
[23]		✓	Multiple	Stationary	Heuristic greedy deep Q-learning	DT-assisted UAV deployment strategy with minimum average delay and hybrid task offloading
[10]	✓		Multiple	✓	Ant colony evolutionary algorithm	Minimize average and peak AoI of sensor nodes
[12]	✓	✓	Single	✓	Decomposition	Propose a high-fidelity DT model with high accuracy between the digital and physical systems by maximizing data collected by all IoT devices
[24]			Multiple	Stationary	Differential Evolutionary Collaborative Optimization	Optimize task offloading decisions, computational resource allocation, and UAV deployment
[15]	✓	✓	Single	✓	Award collection maximization with deep learning	Optimize DT synchronization given energy constraints
[16]	✓	✓			Decomposition with approximation	Optimize DT placement to enhance data freshness
[14]		✓	Multiple	✓	Dueling deep Q-learning with prioritized experience replay	Optimize UAV trajectories in real time under energy constraints
[17]	✓	✓			Status-based and policy-based algorithms	Maximize DT freshness
[18]		✓	Multiple	✓	Scalable Multi-agent DRL (SAMARL, PPO-based, multi-head attention), Digital Twin algorithm	Distributed cooperative target search for AAV swarms; maximize search efficiency, coverage, and safety; sim-to-real validation
[21]		✓	Multiple	Predetermined Trajectory	DRLS, Lyapunov Optimization, Tiny-ML-based Semantic Communications	Minimize DT synchronization latency in UAV-assisted edge computing via optimized semantic extraction and resource allocation
[22]		✓	Multiple	Stationary	Federated Reinforcement Learning (FRL), Deep Q-Network (DQN)	Minimize overall task execution cost (latency & energy) in DT-empowered UAV-assisted MEC by hybrid task offloading and resource allocation with FRL
[25]		✓	Multiple	✓	Behavior-Coupling DDPG (BCDDPG), DT-enabled DRL	Distributed flocking (target-reaching, collision avoidance, connectivity) for multi-UAVs in stochastic environments
[26]		✓	Multiple	✓	DT-enhanced Deep Q-Learning, Q-Learning	Improve RL training speed, convergence, and resource management in dynamic networks
Our Work	✓	✓	Multiple	Stationary	Successive convex approximation; DRL-based approach using TD3	High-fidelity digital twin with practical settings; optimize UAV placement, accuracy, and real-time synchronization

were conducted without deploying any UAVs. Authors in [16] utilized mobile cloud servers for edge computing and only one digital twin of a randomly chosen object was selected to minimize its age at a time, while authors in [17] considered DT update tasks as a part of the optimization process.

To develop practical and scalable solutions, reserach efforts are being done based on reinforcement learning techniques mainly in dynamic environments for UAV-assisted DT-driven frameworks [18], [19], [20], [21], [22]. For example, in [18] a new scalable multi-agent reinforcement learning (MARL) based algorithm is adopted to improve effectiveness and adaptability in a DT-driven framework for multi-UAV cooperative target search applications. Authors in [19] and [20] utilized deep Q-learning in a digital-twin (DT)-assisted task assignment framework in a multi-UAV system to improve dynamic task assignment and resource utilization, and to improve the overall network performance respectively. In [21] authors employed a deep reinforcement learning based synchronization (DRLS) algorithm to generate synchronization actions with low-time complexity in a DT-synchronization framework to minimize the synchronization latency in UAV assisted edge computing environments, while authors in [22] employed federated reinforcement learning to create a generalized RL model to optimize computation offloading and resource allocation in UAV-assisted mobile edge computing (MEC). Table I summarizes recent relevant literature to better highlight the research gap.

In light of the above-surveyed literature, no existing research work, to the best of our knowledge, has jointly optimized accuracy and synchronization neither derived a closed formula for DT synchronization in a multi-UAV-aided DT for an IoT network where multiple sensors are monitoring a single physical process. The main contributions of this work are as follows:

1) We propose a digital twin framework that builds a digital twin with high accuracy and synchronization with the physical system in an IoT network. Every physical entity of the physical system is monitored by a group of IoT devices that upload their measurements to multiple UAVs optimally positioned. The UAVs process all collected measurements to determine the status of the monitored physical entities and deliver this information to the digital twin running at the base station for system monitoring and management.
2) We derive a closed form expression for the novel AoDT metric introduced in [12] and used it to evaluate the synchronization difference between the physical system and its digital counterpart. This will guarantee tackling the challenge of DT real-time synchronization stated earlier.
3) We mathematically formulate the problem as a mixed integer non-convex program to maximize the accuracy of the digital twin by optimizing the amount of collected data while ensuring a target synchronization degree with the physical system.
4) We implement an iterative optimization approach to determine the optimized placement scheme for UAVs

while maximizing data collection and ensuring freshness using successive convex approximations.
5) We develop a reinforcement learning approach based on TD3 to address uncertainty in the problem and handle realistic scenarios with large scale environments by leveraging RL techniques that provide powerful capabilities for making real-time decisions.
6) We present extensive simulation results to evaluate the performance of our proposed solution and demonstrate its effectiveness to build a highly synchronized and accurate digital twin.

III. System Model

We consider a system such as an industrial facility, where several IoT devices are monitoring multiple physical processes with a digital twin running at a nearby base station located at (x_{BS}, y_{BS}, z_{BS}). A set of I IoT devices, denoted by $\mathcal{I} = \{1, 2, \dots, I\}$, is deployed at ground level at $\{x_i, y_i, 0\}$ for monitoring various aspects of the physical environment. IoT devices are supposed to collect real-time measurements, such as temperature, pressure, humidity, vibration, flow rate, and others, of the components of the system, named physical entities. Due to their limited transmission power and lack of processing capabilities, IoT devices deliver their readings to a set of J stationary UAVs denoted by $\mathcal{J} = \{1, 2, \dots, J\}$, and that are optimally positioned at the coordinates (x_j, y_j, H), $\forall j \in \mathcal{J}$ where H is the height UAVs are positioned. UAV repositioning can be performed based on network demand between mission intervals, allowing flexibility when operational conditions permit. The UAVs are assumed to be fully meshed with a high-speed link and they use their computational power to process the measurements and determine the current status of the physical entities. Status updates of the physical entities are then uploaded to the digital twin on the BS in a way to ensure high synchronization with the physical system. Fig. 1 illustrates this system model.

Depending on the size, structure, or functionality of each physical entity, multiple IoT devices may be needed to capture measurements from different angles or locations. Each IoT device i generates tasks of size S_i in bytes that are uploaded to UAV j. However, if UAV j lacks sufficient computational resources to process a task, it forwards the task to another UAV k for processing. UAVs are considered to operate in full duplex mode, allowing them to receive and transmit simultaneously, while using orthogonal channels to avoid self-interference. Moreover, UAVs opt to use the most recent measurements coming from IoTs to ensure that the resulting status reliably mirrors the current status of the monitored physical entity. This being said, a UAV discards old data once it obtains fresh measurements from a respective IoT.

A. Communication Channel Model

Ground-to-air (G2A) communication between UAVs and ground IoT devices is based on a probabilistic path loss model that accounts for line-of-sight (LoS) and non-line-of-sight (NLoS) components. Accordingly, we employ the following LoS and NLoS channel models between IoT device i and UAV j [23]:

$$L_{i,j}^{LoS} = J_{FS} + 20 \log d_{i,j} + \eta_{LoS}, \tag{1}$$

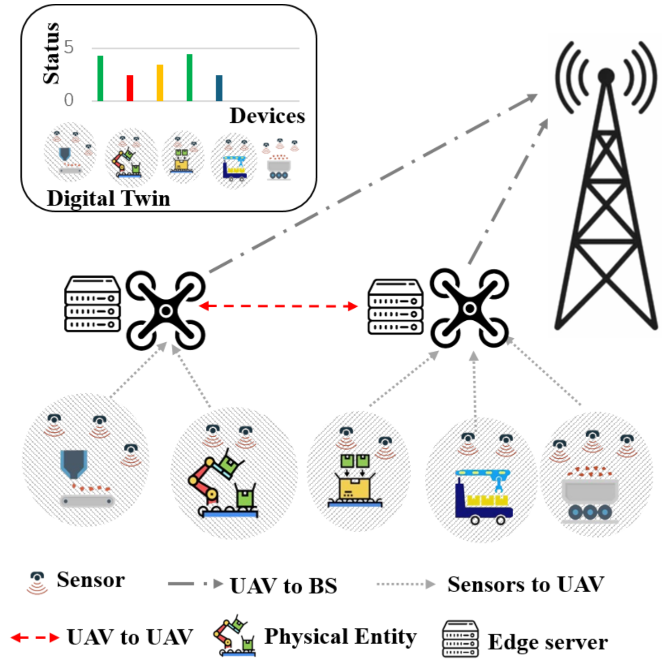

Fig. 1. System model.

and

$$L_{i,j}^{NLoS} = J_{FS} + 20 \log d_{i,j} + \eta_{NLoS}, \tag{2}$$

where $\mathrm{J}_{FS} = 20 \log f_c + 20 \log \frac{4\pi}{c}$, the variable f_c signifies the system's carrier frequency, and c is the speed of light's constant. The parameters η_{LoS} and η_{NLoS} are additional attenuation factors specific to the line-of-sight (LoS) and non-line-of-sight (NLoS) link environments, respectively. $d_{i,j}$ denotes the Euclidean distance between device i and UAV j is represented by:

$$d_{i,j} = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2 + H^2}. \tag{3}$$

In addition, the probability of a LoS communication link, denoted as $P_{i,j}^{LoS}$, between a UAV j an IoT i device is represented as:

$$P_{i,j}^{LoS} = \frac{1}{1 + a \cdot \exp\left(-b \cdot \left(\arcsin\left(\frac{z_j}{d_{i,j}}\right) - a\right)\right)} \tag{4}$$

where a and b are environment parameters. As a result, the average path loss is expressed to be:

$$L_{i,j}^{avg} = P_{i,j}^{LoS} L_{ij}^{LoS} + (1 - P_{i,j}^{LoS}) L_{i,j}^{NLoS} \tag{5}$$

Based on the above, the achieved uplink rate from IoT i to UAV j is computed as [23]:

$$r_{i,j} = B_{ij} \log_2 \left(1 + p_i \cdot 10^{\frac{-L_{i,j}^{avg}}{10}} \cdot \frac{1}{\sigma^2}\right) \tag{6}$$

where B_{ij} represents the channel bandwidth that is distributed among the devices based on their needs, p_i is the power level of IoT device i, and σ^2 is the Gaussian white noise power.

For simplicity, we assume a high-speed LoS link between UAVs. Therefore, the elapsed time to upload data from the associated UAV j with the IoT device i to another UAV k is considered constant and equal to T_{u2u}. The UAVs are assumed to communicate with the base station via high capacity fronthaul links (such as free space optics (FSO) or millimeter-wave (mmWave) links) with high communication speed and low latency [27], [28]. According to related literature [29], [30], the download time of the computed results from the UAV to the BS can be neglected, since the processed data is relatively small.

B. Data Generation and Computing Models

During monitoring a physical entity, each IoT device i generates data based on a Poisson distribution [31], [32] with an average rate λ_i. The data generated from multiple IoT devices is transferred to UAV j for potential processing.

Since each associated IoT device generates its tasks according to a Poisson process with rate λ_i, and assuming a constant propagation delay from the IoT device i to UAV j, the aggregated workload from a set $\mathcal{N}_j$ of IoT devices on UAV j can also be modeled as a Poisson process with rate $\sum_{k\in\mathcal{N}_j}\lambda_k$.

We assume that the service times at UAV j are independent and identically distributed exponential random variables, with an average service time of $\frac{1}{\mu_j}$, where μ_j resembles the average service rate of the UAV j. The service rate μ_j, measured in requests per second, can be expressed as:

$$\mu_j = \frac{f_j}{L} \tag{7}$$

where f_j is the processing capacity of UAV j in cycles per second, and L is the average task size in cycles. Hence, we model each UAV j as an M / M / 1 queueing system with an arrival rate $\lambda_{total,j} = \sum_{k\in\mathcal{N}_j}\lambda_k$ and a service rate μ_j [33], [34].

We define the offered load ρ_i from IoT device i to the processing UAV j by:

$$\rho_i = \frac{\lambda_i}{\mu_j} \tag{8}$$

Thus, the total offered load at UAV j where the tasks from a set of $\mathcal{N}_j$ IoT devices are processed is:

$$\rho_{total,j} = \frac{\lambda_{total,j}}{\mu_j} = \frac{\sum\limits_{k\in\mathcal{N}_j}\lambda_k}{\mu_j} \tag{9}$$

C. Modeling Quality of Service in a Digital Twin

Many research is emerging to define methods and metrics for evaluating digital twin performance. Authors in [35] illustrated that ensuring real-time synchronization between a DT and its physical counterpart as well as maintaining high accuracy in a DT is crucial for effective DT adoption in different applications. Authors in [36] introduced digital twin entanglement metric that represents to which extent the digital and physical systems' synchronization process fulfills the needs of a specific application. Authors took into account the freshness of the collected data and the ratio of collected to total data. This ratio reflects the DT accuracy. Moreover, authors in [37] illustrated that DT freshness may be defined using the age of information concept as the elapsed time after a DT has been updated. However, they did not derive any closed formula to model this metric. DT freshness can be best represented through a novel metric introduced in [12] as age of digital twin metric (AoDT) that takes into account: (i) time of data upload from sensors to edge device; (ii) time of data processing to reflect device status; and (iii) time of data download to BS for DT updates. Authors in [12] considered a simple scenario where one IoT monitors one physical process. They proposed a problem to optimize DT accuracy subject to AoDT constraints. However; in practice, multiple IoT devices might monitor the same physical entity. In this work, we formulate a closed form expression to model age of digital twin in such a practical scenario.

The age of information (AoI) is a destination-centric measure for information freshness that counts the time that has passed since the last fresh update was generated at the source. In this study, we apply the concept of AoI to the setting of digital twins, where we need to express the freshness of the physical entity's status through the responsive DT. As a result, we utilize the Age of Digital Twin (AoDT) metric introduced in [12] and defined as the elapsed time between data generation and the status update received by the BS.

The calculation of the AoDT for each monitored process accounts for different phases after data collection, namely:

- Upload phase: The data is transmitted to the associated UAV through the wireless channel for processing, and then forwarded from the associated UAV to another one when the first UAV lacks computation resources.
- Processing phase: The uploaded data enters the computation queue at the UAV and is then processed to obtain results.
- Download phase: The data is processed from the UAV to the BS via the LoS channel to update the DT's status.

We define the instantaneous AoDT at time t for the ith IoT device as follows:

$$\zeta_i(t) = t - u_i(t) \tag{10}$$

where $u_i(t)$ is the instant generation of the last update of the DT from IoT i. Figure. 2 illustrates an example of the evolution of the AoDT $\zeta(t)$ for IoT device i over time t. Without loss of generality, suppose we start monitoring at time $t = 0$, when the computation queue is empty and the age is $\zeta(0)$. In the absence of updates, the AoDT for the DT corresponding to the source i climbs linearly with time and is reset to a lower value when an update is received.

Update l for IoT device i, generated at time t_l^g, is uploaded to the associated UAV j and then to another UAV k if the UAV j lacks computation resources. We denote by t_l^u the upload time. As a result, we can express the upload duration $X_l = t_l^u - t_l^g$ of task l for IoT device i as follows:

$$X_l = \begin{cases} \dfrac{S_i}{r_{ij}} & \text{if the task } l \text{is processed} \\ & \text{by the associated UAV} j \\ \dfrac{S_i}{r_{ij}} + T_{u2u} & \text{otherwise} \end{cases} \tag{11}$$

where S_i represents the generated task size in bytes from IoT device i, and T_{u2u} is the required time to upload the generated data of the device i from its associated UAV j to another one, when j lacks sufficient computation resources.

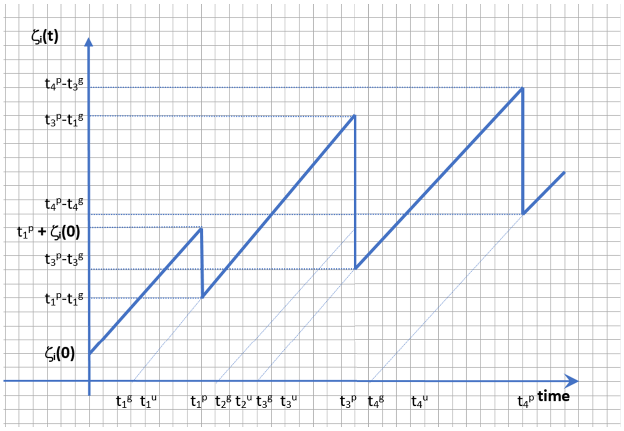

Fig. 2. Evolution of AoDT example.

Assuming stable radio conditions for each IoT device i with tasks of the same size S_i, the upload time X_l is constant and equal to D_i for all l.

After the uploading phase, data is processed at t_l^p, and the result is received by the BS to update the status of the corresponding DT at t_l^d. At t_l^d, the AoDT $\zeta_i(t_l^d)$ is reset to $(t_l^d - t_l^g)$.

Regarding the download phase of the processed task l from UAVs to the BS, the download duration is given by:

$$Z_l = t_l^d - t_l^p \tag{12}$$

Therefore, the evolution of the AoDT is mainly affected by the processing phase since the time elapsed at other phases is considered constant. Given that we have modeled every UAV j as a M/M/1 system and infinite buffer, we can select either first-come, first-serve (FCFS) or last-come, first-serve (LCFS) queuing discipline.

Noting that the FCFS queue paradigm allows new update messages to be queued behind outdated messages issued earlier, and that we want to transfer the most recent data from IoT to reduce the AoDT, it is preferable to use the LCFS queuing discipline. Considering LCFS, fresh information from an IoT device preempts any previously queued update packets, and the preempted data is deleted. Furthermore, based on the performance evaluation in [38], the AoI values achieved using the LCFS strategy outperformed those obtained under the FCFS strategy. Hence, we will consider LCFS with preemption in service (LCFS-S). When a new packet is uploaded, it preempts the packets that are being served. We can show from Fig. 2 after the third generated update task at time t_3^g is uploaded to the UAV at time t_3^u, it preempts the previous task in service to process recent data.

For the proposed system, multiple IoT devices monitor a single physical process. Each device $i \in N_k$, where N_k represents the set of devices monitoring the same process, generates updates at a rate λ_i following a Poisson process. Updates are transmitted directly to a UAV without an aggregator. Then, the UAV processes each update and updates the state of the corresponding Digital Twin (DT). Once an update is processed, the Age of Digital Twin (AoDT) for the corresponding DT is reset.

Since multiple IoT devices directly send updates to the UAV, the min rate should be taken into consideration to wait for the slowest update to be received, λ_{N_k} is given as:

$$\lambda_{N_k} = \min_{i \in \mathcal{N}_k} \lambda_i, \tag{13}$$

The Average Age of Information (AAoI) for a multiple-source, single-server system with LCFS preemption in service [38] is:

$$\Delta = \frac{1}{\lambda}\left(1 + \frac{\lambda}{\mu}\right), \tag{14}$$

where λ is the total arrival rate and μ is the service rate.

For our system, the average arrival rate λ_{N_k} replaces λ. Hence, the average AoDT for a Digital Twin DT_k monitoring the physical process is:

$$\Delta_k = D_{N_k} + \frac{1}{\lambda_{N_k}}\left(1 + \frac{\lambda_{N_k}}{\mu}\right), \tag{15}$$

where D_{N_k} represents the maximum upload time for updates from the devices in N_k and is given as:

$$D_{N_k} = \max_{i \in N_k} D_i, \tag{16}$$

where D_i is the upload time for device i.

Substituting D_{N_k} into the AoDT expression, we obtain the final form of the average AoDT:

$$\Delta_{DT_k} = \max_{i \in N_k} D_i + \frac{1}{\lambda_{N_k}}\left(1 + \frac{\sum_{i \in N_k} \lambda_i}{\mu}\right). \tag{17}$$

IV. Problem Formulation

In this section, we formulate a non-convex optimization problem ($\mathcal{P}$) to maximize the DT accuracy. We consider the scenario in which multiple IoT devices monitor a single physical entity. Each IoT device $i \in \mathcal{I}$ is associated with a single UAV $j \in \mathcal{J}$ that receives uploaded tasks and processes them. However, when the associated UAV has limited computational capabilities, it will delegate the tasks to another UAV for processing. We define two binary decision variables a_{ij} and b_{ij}. The first variable a_{ij} indicates whether IoT i is associated with UAV j.

$$a_{ij} = \begin{cases} 1 & \text{if IoT device } i \text{is associated with UAV} j \\ 0 & \text{otherwise} \end{cases} \tag{18}$$

The second binary variable b_{ij} indicates whether the tasks of IoT i are processed by UAV j.

$$b_{ij} = \begin{cases} 1 & \text{if the tasks generated by IoT device } i \\ & \text{are processed by UAV} j \\ 0 & \text{otherwise} \end{cases} \tag{19}$$

Since multiple sensors are monitoring one process, we define $\mathcal{K}$ groups or sets of IoT sensors where each set N_k is monitoring a process as stated earlier.

We use δ_{ik} to specify the set to which the IoT device belongs.

$$\delta_{ik} = \begin{cases} 1, & \text{if IoT device } i \text{is assigned to a set } k \\ 0, & \text{otherwise} \end{cases}$$

The formulated problem ($\mathcal{P}$) determines the optimized position for each UAV $j \in \mathcal{J}$. Furthermore, ($\mathcal{P}$) derives for each IoT device $i \in \mathcal{I}$ its associated UAV and the UAV responsible for processing its tasks.

$$(\mathcal{P}) : \max \sum_{i=1}^{I} \sum_{j=1}^{J} a_{ij} r_{ij} \quad (20)$$

$$\text{s.t.} \quad \sum_{j=1}^{J} a_{ij} = 1, \quad \forall i \in N_k, \quad (21)$$

$$\sum_{j=1}^{J} b_{ij}^k = \delta_{ik}, \quad \forall i \in N_k, k \in \mathcal{K}, \quad (22)$$

$$\delta_{ik}.\delta_{nk}.b_{ij}^k = \delta_{ik}.\delta_{nk}.b_{nj}^k, \quad \forall i, n \in N_k, j \in \mathcal{J}, k \in \mathcal{K}, \quad (23)$$

$$\frac{f_j}{L} - \sum_{i=k}^{K} \sum_{i=1}^{I} b_{ij}^k \lambda_i \geq 0, \quad \forall j \in \mathcal{J}, \quad (24)$$

$$r_{ij} \geq a_{ij} R_{\min}, \quad \forall i \in N_k, j \in \mathcal{J}, \quad (25)$$

$$B_{ij} \leq a_{ij} \cdot M, \quad \forall i \in N_k, j \in \mathcal{J}, M \to \infty \quad (26)$$

$$\sum_{i=1}^{I} \sum_{j=1}^{J} B_{ij} \leq B_{\text{sys}}, \quad (27)$$

$$\sqrt{(x_j - x_l)^2 + (y_j - y_l)^2 + (z_j - z_l)^2} \geq \theta, \quad \forall j, l \in \mathcal{J}, \quad (28)$$

$$\lambda_{N_k} \leq \lambda_i, \quad \forall i \in N_k, \quad (29)$$

$$\text{v}^k \leq T_k, \quad \forall k \in \mathcal{K}, \quad (30)$$

$$\text{v}^k \geq \sum_{j=1}^{J} a_{ij} \frac{S_i}{r_{ij}} + \frac{1}{\lambda_{N_k}} \left(1 + \frac{\sum_{i \in N_k} \lambda_i}{\mu}\right) + \left(1 - \sum_{j=1}^{J} a_{ij} b_{ij}^k\right) \cdot T_{\text{u2u}} \quad \forall i \in N_k, k \in \mathcal{K}, \quad (31)$$

Problem $(\mathcal{P})$ optimizes data collection from IoT devices by maximizing the sum rate while considering constraints related to computation tasks, AoDT, available resources, and overall system performance. The objective function (20) aims to maximize the sum rate. Constraints (21) and (22) ensure that each IoT device i is associated with a single UAV j and that the tasks of each device i are being processed by a single UAV j. Constraint (24) is crucial to maintaining a stable queuing system where we ensure that the rate at which computational requests arrive is lower than the service rate for each UAV. To guarantee a minimum quality of service (QoS), constraint (25) ensures that the communication rate between the IoT devices and the UAV is higher than the target value. Constraint (26) ensures that the bandwidth is set to zero if there is no association between IoT i and UAV j. Constraint (27) ensures that the total uplink bandwidth of the system does not exceed the available system bandwidth of value B_{sys}. Constraint (28) mandates a minimum safety distance θ between any two deployed UAVs. Constraints (30) and (31) ensure that the average AoDT Δ_k for IoT device i remains below a predefined threshold T_K to maintain the freshness of data updates.

Algorithm 1 SCA-Based Solution of Problem $\mathcal{P}$

Input: $\mathcal{I}$, $\mathcal{J}$, positions of IoT devices, position of the base station, error tolerance

Initialization: Set initial position for UAVs, bandwidth, and iteration number n

while $|\text{Obj}\mathcal{P}(n) - \text{Obj}\mathcal{P}(n-1)| \leq \epsilon$ **do**

  Solve the convex problem $\mathcal{P}$ to obtain optimal UAV positions and system bandwidth

  Update UAV positions

  Update bandwidth value

  Update sum rate matrix

  Update $n = n + 1$

end while

Output: The optimal UAV positioning and the maximum system bandwidth

V. Optimization Solution Approach

Problem $(\mathcal{P})$ is a mixed integer non-convex optimization problem. In this section, introduce a successive convex approximation (SCA) approach to solve the problem and determine the optimized positions of the UAVs in a way to maximize the data collection rates under AoDT threshold to improve both accuracy and synchronization of the DT. Constraint 25 is non-convex and constraints (3), (4), (5) are nonlinear. These constraints are linearized using first-order Taylor approximations. Algorithm (1) presents the successive convex approximations of $(\mathcal{P})$.

We used the Matlab Mosek cvx solver, which applies a primal-dual interior-point algorithm that simultaneously solves the primal and dual problems to achieve good accuracy. This solver iteratively solves continuous convex optimization problems by moving within the feasible region while maximizing the objective function. To keep variables within the feasible region, interior-point methods work by using Newton's method on "barrier" functions. Also, Mosek's primal-dual interior-point method ensures fast convergence for the high complexity of the problem involving UAV placement and data collection. Thus, the worst-case complexity of this problem can be roughly estimated as $\text{C} \sim \sqrt{\text{number of variables}}$ because the complexity depends on the number of Newton steps, in which the latter is influenced by the number of variables of the optimization problem and the iterations needed for the algorithm to converge given an initial start as stated in [12].

VI. Twin Delayed Deep Deterministic Policy Gradient-Based Solution Approach

In UAV-assisted IoT networks with DT functionality, real-time decision-making and dynamic optimization are crucial in maximizing data collection efficiency while maintaining AoDT constraints. Owing to the complexity of the optimization problem due to its nonconvex constraints and to cope with uncertainty and to ensure real time adaptability especially in dynamic environments, we resort to deep reinforcement learning (DRL) as a viable solution approach [23].

The standard DDPG algorithm occasionally encounters instability and hyperparameter sensitivity, due to Q-value overestimation, and hence produces a non-optimal or divergent policy in highly dynamic and continuous decision-making scenarios. Therefore, in our work, we adopt the TD3 algorithm, an off-policy actor-critic method that extends the DDPG framework to continuous action spaces. It learns a deterministic policy using two Q-value critics to estimate the optimal policy value while addressing value-function overestimation bias. This is achieved through three key techniques: clipped Double Q-learning, delayed policy updates, and target policy smoothing. The most crucial one is the clipped Double Q-Learning technique, which trains two independent Q-networks in parallel and uses the lower estimate as the target value to enhance stability and robustness. The thing that distinguishes TD3 from conventional DDPG algorithms and addresses their limitations in overestimation bias and training instability [39], [40]. Combining the three key ideas of TD3, our approach mitigates these problems, achieving stable training dynamics and improved performance. Based on these improvements, the algorithm's design aligns with our research objectives, enabling dynamic optimization of UAV positioning and resource allocation, while actively resorting to the specified system constraints, mainly AoDT constraints, and channel conditions.

A. MDP Formulation

We formulate the problem as an MDP problem that is formally defined by the tuple (S, A, R, P, γ), where

- S is the set of all states $s \in S$,
- A is the set of all actions $a \in A$,
- $R(s, a)$ denotes the immediate reward obtained by taking action a in state s
- P is the probability of shifting from one state to another given an action
- γ is the discount factor that defines the action to take in each state

A DRL approach based on the TD3 algorithm is proposed for finding the policy that optimizes UAV positioning, IoT-to-UAV assignment, and edge-computation scheduling under unknown activation patterns. We customize TD3 to our framework by redefining the state S, action A, and reward R functions as follows:

- **State:** The state space of our TD3-based reinforcement learning algorithm is defined by the initial positions of UAVs and IoT devices, task arrival rate, allocated bandwidth for the whole system, current load on UAV j, average AoDT, and communication channel conditions. Thus, the state at time step t is represented as:

$$S(t) = \begin{Bmatrix} (x_j(t), y_j(t), H), (x_i, y_i), \Delta_{DT}(t), \\ B_{sys}(t), L_j(t), \lambda_i(t) \end{Bmatrix} \tag{32}$$

- **Action:** The action space at each time step t involves the continuous adjustment of UAV positions to maximize the sum rate, while having the best association, computation assignments, and bandwidth allocation between IoTs and UAVs. We define our continuous action space as:

$$A(t) = \{(\Delta x_j(t), \Delta y_j(t)), a_{ij}(t), b_{ij}(t), B_{ij}(t)\} \tag{33}$$

where $(\Delta x_j(t), \Delta y_j(t))$ represents positional adjustments for each UAV j at a fixed height.

- **Reward:** The reward function $r(t)$ simultaneously maximizes the sum rate collected by UAVs with penalties for violating AoDT constraints, minimum safety distance between any two UAVs, and any other previously defined constraints. The reward at time step t is defined as:

$$\begin{aligned} r(t) = \alpha \sum_{i,j} a_{ij}(t)\, r_{ij}(t) - \beta \sum_{k} \max(0, \Delta_{DT_k}(t) - T_k) \\ - \gamma P_{dist}(t) - \delta V_{viol}(t) \end{aligned} \tag{34}$$

where $\sum_{i,j} a_{ij}(t) \cdot r_{ij}(t)$ represents the total sum rate collected by UAVs at time t, while $\sum_k \max(0, \Delta_{DT_k}(t) - T_k)$ captures the AoDT violation penalties, where T_k is the AoDT threshold. $P_{dist}(t)$ penalizes any pair of UAVs closer than θ, and $V_{viol}(t)$ accounts for any breach of QoS, CPU, or bandwidth restriction. In our implementation, the weighting coefficients α, β, γ and δ are set through scaling in which the sum rate is normalized and the AoDT penalty is given the highest weight to reflect its critical impact on system performance. This ensures that our research objectives focus on optimizing UAV positions and resource allocation dynamically in response to varying IoT demands, channel conditions, and AoDT requirements.

Fig. 3 presents our system architecture, divided into three interconnected components. On the left, the RL Module includes the actor-critic pair: the actor proposes continuous control commands (UAV motion deltas, IoT-UAV associations and bandwidth allocations) which are evaluated by two critic networks. Then, the experience tuples are stored in a replay buffer and used, along with clipped target actions from slowly updated target networks, to perform off-policy learning. In the middle, the digital model simulates the environment state, including UAV and IoT locations, task arrival rates, allocated resources and resulting communication metrics, and computes the scalar reward by aggregating throughput, AoDT penalties, distance penalties, and constraint violations. Finally, on the right, the physical entities depict the UAV fleet and IoT devices executing the digital model's commands and feeding back sensor data, thus completing the cycle from perception to action.

B. RL Algorithm

In this subsection, we delineate our TD3-based solution presented in Algorithm 2 to optimize UAV placement and task offloading under AoDT constraints in a continuously evolving environment. We train a TD3 agent where at each time step t the agent observes a high-dimensional state s_t (including UAV/IoT positions, group assignments, loads, AoDT values, and bandwidths) and selects a continuous action a_t (UAV positioning plus IoT-to-UAV association and processing logits with added exploration noise). The environment parses a_t, updates UAV positions, recomputes bandwidth, data rates, and AoDT penalties, and returns a scalar reward r_t that balances normalized throughput against penalties for AoDT threshold violations, inter-UAV proximity breaches, and other constraint violations. Each transition (s_t, a_t, r_t, s_{t+1}) is stored in a replay buffer. On every step, both critic networks are updated via Bellman backups using clipped Double Q targets,

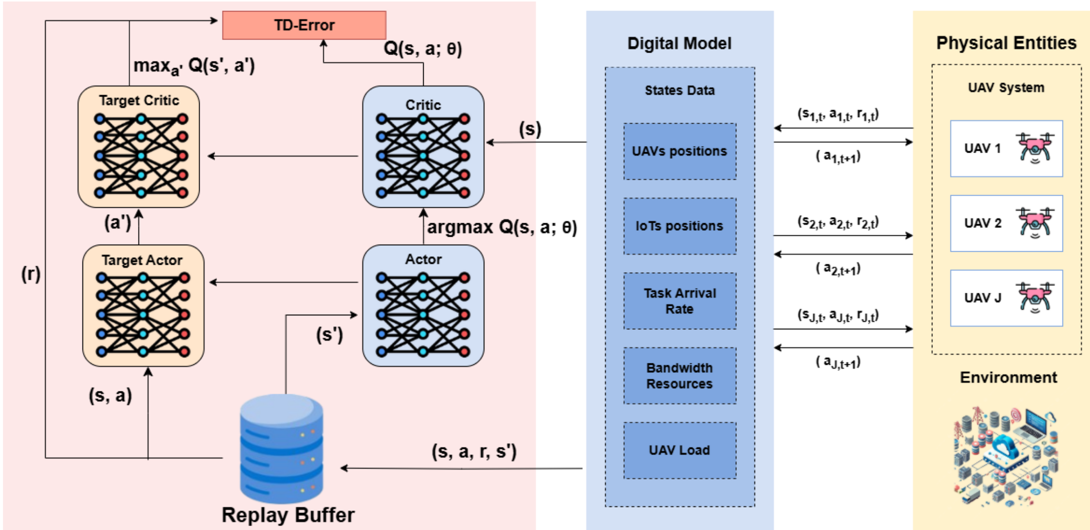

Fig. 3. TD3 solution framework.

TABLE II
SIMULATION PARAMETERS

Parameter	Value
Minimum data rate, R_{min}	10000 bits/sec
Minimum bandwidth allocation, B_{sys}	20000 Hz
Distance limit between UAVs, θ	10 m
UAV altitude, z_{UAV}	100 m
Upload time between UAVs, T_{u2u}	0.3 sec
Carrier frequency, f_c	1×10^6 Hz
Task arrival rate, λ_i	2 tasks/sec
UAV processing frequency, f_j	2×10^8 Hz
Speed of light, c	3×10^8 m/s
AoDT threshold, T_K	2.8 sec
LoS additional loss, η_{LoS}	1 (linear scale)
NLoS additional loss, η_{NLoS}	21 (linear scale)
Noise power, σ	10×10^{-3} W
Power, pi	0.2 W
Environment parameter, a	9.61
Environment parameter, b	0.16

and once every d steps the actor network and all target networks are softly updated. This iterative process continues until the policy converges, yielding a trained TD3 agent capable of dynamically optimizing UAV deployment strategies under resource and freshness constraints in digital twin-enhanced aerial computing networks. Regarding complexity, the computational complexity of the TD3 algorithm depends on its actor and critic networks. Assuming m denotes the number of parameters in each critic network, and n denotes the number of parameters in the actor network, B the mini-batch size, and T the total number of training steps, then the overall training complexity of TD3 scales as $O\left(T\left(mB + \frac{nB}{d}\right)\right)$. Therefore, TD3 has linear time complexity in the number of training steps and linear memory complexity in the size of stored transitions.

VII. SIMULATION RESULTS AND PERFORMANCE ANALYSIS

In this section, we evaluate the proposed solution approach formulated above. We consider an industrial field of $500 \times 500\,\text{m}^2$ where ten IoT devices are randomly deployed, with every five devices monitoring one process. We consider the system parameters as shown in Table II unless otherwise stated. We present different simulation results of the TD3-based solution compared to the SCA-based solution and two baseline approaches, k-means and random placement of UAVs as an initialization step. The k-means positioning approach determines UAV locations by minimizing the sum of distances between the IoT devices and their cluster centroids where the UAV is supposed to be positioned. In contrast, the random deployment method assigns UAV positions without any specific strategy, disregarding the locations of IoT devices or any optimization concerns.

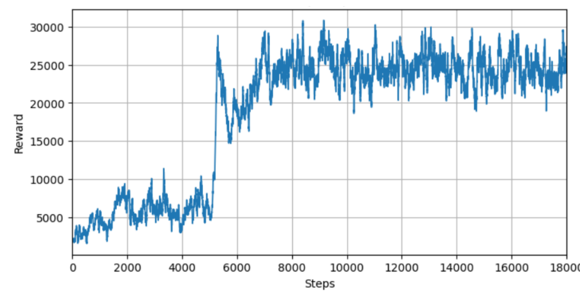

Fig. 4. Convergence of the TD-3 solution approach.

For simulation, we used DELL laptop 13^{th} Gen Intel(R) Core(TM) i7-1355U, an x64-based processor, and a CPU core running at 2.06 GHz with 16 GB RAM. Note that each data point in the figures represents the average of 20 different random runs. In Figure 4, we demonstrate the convergence characteristic of the proposed TD3-based solution. The figure shows that the learning curve for cumulative reward increases as the agent starts to explore the environment and change the position of the UAVs based on the penalties imposed on AoDT and the task arrival rates. The curve starts to stabilize upon further training, which proves that the TD3 agent has learned the best policy on how to balance UAV trajectory and bandwidth allocation while satisfying the AoDT constraint. The agent reaches a stable behavior after approximately 7000 steps. This convergence curve demonstrates a high degree of the TD3 agent's generalization ability in the Digital Twin

Algorithm 2 TD3-Based Solution

Require: Environment $\mathcal{E} \leftarrow \texttt{UAVAoDTEnv}(N_u, N_i, \text{area}, K)$
  Actor network μ_ϕ, critics $Q_{\theta_1}(s,a), Q_{\theta_2}(s,a)$ and targets $\mu_{\phi'}(s), Q_{\theta_1'}(s,a), Q_{\theta_2'}(s,a)$
  Replay buffer $\mathcal{D}$, noise scale σ, policy delay d, discount γ, soft-update rate τ, total steps T
  Initialize ϕ, θ_1, θ_2 and set $\phi' \leftarrow \phi, \theta_1' \leftarrow \theta_1, \theta_2' \leftarrow \theta_2$
  Initialize replay buffer $\mathcal{D} \leftarrow \emptyset$
  for $t = 1$**to**T **do**
    Select action $a_t = \mu_\phi(s_t) + \epsilon$, with noise $\epsilon \sim \mathcal{N}(0, \sigma)$
    Execute action a_t and observe reward r_{t+1} and next state s_{t+1}:
  -parse a_t into movement deltas, association logits, processing logits
  -update UAV positions: $\Delta x, \Delta y \times 10$, clip to $[0, \text{area}]$
  - $a_{ij} \leftarrow \texttt{normalize\_association}(\textit{associate}\texttt{\_logits})$
  - $b_{ij} \leftarrow \texttt{normalize\_processing}(\textit{process}\texttt{\_logits})$
  `-update bandwidth, calculate rates and loads, calculate AoDT`
  -compute sum rate, AoDT penalty, dist penalty, violations
  -compute reward $r = \frac{\text{sum rate}}{R_{\max}} - 10\, P_{\text{AoDT}} - 5\, P_{\text{dist}} - 5\, V_{\text{viol}}$
  -$s_{t+1} \leftarrow \mathcal{E}.\texttt{get\_obs}()$
    Store $(s_t, a_t, r_{t+1}, s_{t+1})$ in replay buffer $\mathcal{D}$
    if enough samples in replay buffer $\mathcal{D}$ & it's time to update **then**
      Sample minibatch $\{(s_i, a_i, r_i, s_{i+1})\}_{i=1}^N$ from $\mathcal{D}$
      Compute target actions with noise:
  $a_{i+1}' = \mu_{\phi'}(s_{i+1}) + \epsilon'$, with $\epsilon' \sim \text{clip}(\mathcal{N}(0, \sigma'), -c, c)$
      Compute target Q-value:
  $y_i = r_i + \gamma \, \min_{j=1,2} Q_{\theta_j'}(s_{i+1}, a_{i+1}')$
      Update each critic
  $\theta_j \leftarrow \theta_j - \eta \, \nabla_{\theta_j} \frac{1}{N} \sum_i \left[Q_{\theta_j}(s_i, a_i) - y_i\right]^2$
      if $t \bmod d = 0$ **then**
        Update actor:
  $\phi \leftarrow \phi + \eta \, \nabla_\phi \frac{1}{N} \sum_i Q_{\theta_1}(s_i, \mu_\phi(s_i))$
        Soft-update targets:
        for $j \in \{1, 2\}$ **do**
          $\theta_j' \leftarrow \tau \, \theta_j + (1-\tau) \, \theta_j'$
        end for
        $\phi' \leftarrow \tau \, \phi + (1-\tau) \, \phi'$
      end if
    end if
    $s \leftarrow s_{t+1}$
  end for

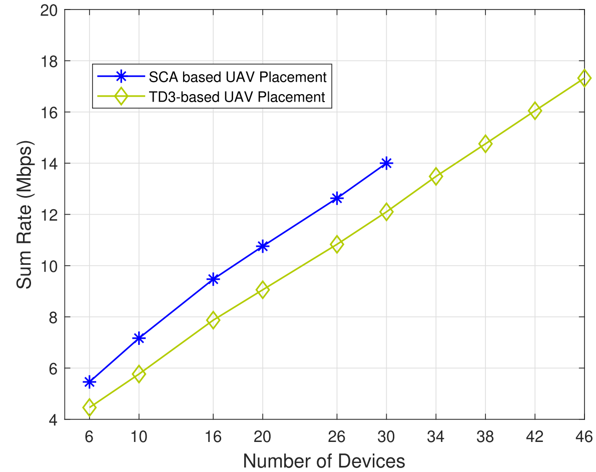

Fig. 5. Peformance of TD3-based solution versus SCA-based solution.

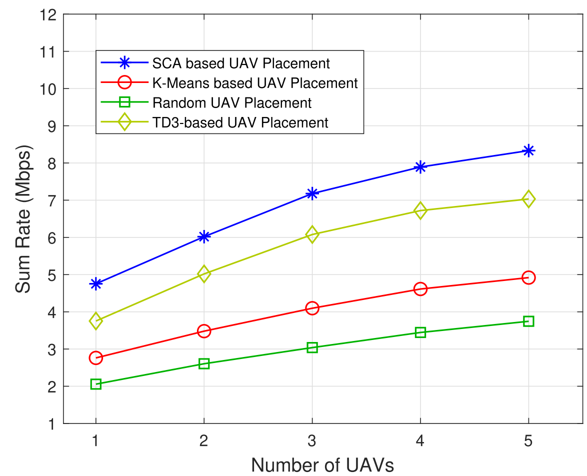

Fig. 6. Sum rate vs. number of UAVs, where $\mathcal{I} = 10$.

environment, as it can identify and execute accurately the movements and offloading actions that lead to long-term reward optimization when the synchronization and resource constraints are applied.

In Fig. 5, we demonstrate the scalability property of the proposed TD3-based solution as compared to SCA-based approach. For larger network sizes, the SCA-based solution fails to give results in reasonable time, exceeding two hours with 30 IoT devices. As we increase the number of devices, the TD3-based solution continues to provide highly optimized solutions within few minutes.

Fig. 6 shows the relationship between the number of deployed UAVs and the sum rate. The sum rate improves across all four approaches as more UAVs are deployed due to enhanced resource utilization and spatial coverage, enabling efficient data collection to ensure fresh status updates for the DT. SCA-based approach consistently outperforms all other approaches, achieving the highest sum rate of approximately 8.8 Mbps with the deployment of five UAVs, compared to 7 Mbps for TD3-based placement, 5.6 Mbps for k-means placement, and 3.4 Mbps for random placement. These results illustrate that SCA-based approach achieves the best performance, however, at the expense of running time as demonstrated in the earlier figure. TD3-based solution achieves highly optimized results notably exceeding the two baseline approaches. It ensures timely updates, improving synchronization for the digital twin, which, in turn, enhances data freshness and accuracy of the digital model.

Increasing the number of IoT devices, Fig 7 shows an increase in the sum rate for a fixed number of deployed UAVs (three), across all placement strategies. This reflects the increased data generation effect and improved utilization of network resources for parallel and efficient data collection by UAVs. The SCA-based solution approach achieved a maximum sum rate exceeding approximately 14 Mbps for

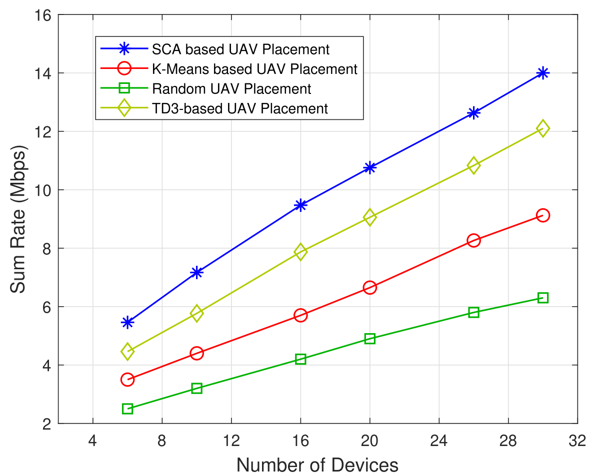

Fig. 7. Sum rate in terms of IoT device count where $\mathcal{J} = 3$.

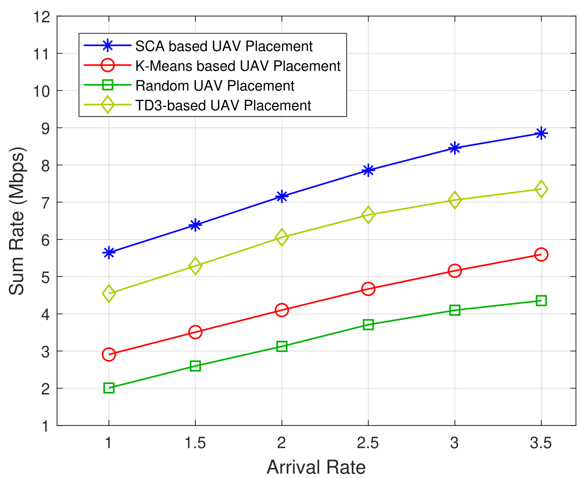

Fig. 8. Sum rate vs. task arrival rate, with $\mathcal{I} = 10$ and $\mathcal{J} = 3$.

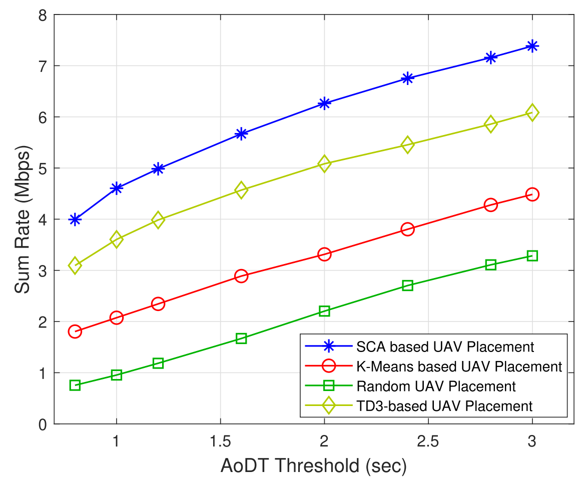

Fig. 9. Sum rate in terms of age of digital twin threshold, with $\mathcal{I} = 10$ and $\mathcal{J} = 3$.

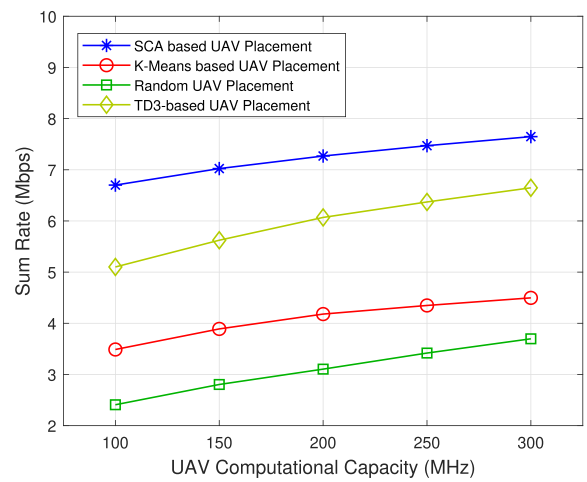

Fig. 10. Sum rate in terms of UAV's computational capacity, with $\mathcal{I} = 10$ and $\mathcal{J} = 3$.

32 devices, followed by the TD3-based solution that remains superior to the baseline approaches. TD3 solution achieved 11.6 Mbps for 32 devices, while the two baseline approaches achieved 7.8 Mbps, and 5.2 Mbps, respectively. This considerable performance gap underscores the effectiveness of our proposed TD3 solution that maintains a highly synchronized and accurate digital twin with low complexity.

In Fig 8 we increase the task arrival rate from 1 to 3.5 tasks per second and the achieved sum rate improves across all UAV placement methods. However, TD3-based placement shows consistent performance while increasing the arrival rate and remains closest to the SCA-based solution. It achieves approximately 7.3 Mbps, while K-means placement achieves 5.3 Mbps and random placement achieves 4.2 Mbps for a task arrival rate of 3.5 per each device. These results demonstrate how a higher arrival rate affects the total sum rate, as more data packets are generated and transmitted.

To better demonstrate the impact of the AoDT metric on the system performance, we compared the total sum rate in Fig. 9 while varying the AoDT threshold, starting from 0.8 seconds, across different UAV placement strategies. All placement approaches demonstrate an increase in sum rate with a more relaxed AoDT threshold. The TD3-based solution was the closest to the SCA-based results, reaching 6 Mbps compared to k-means, and random UAV placement that achieved around 4.4 Mbps, and 3.2 Mbps, respectively, when the threshold was relaxed to 3 seconds. The performance of the baseline approaches highlights their limitations in improving system efficiency due to poor spatial configuration, resulting in the under-utilization of available resources. In contrast, the TD3-based placement significantly enhances system efficiency and network throughput, particularly as the AoDT threshold is relaxed, allowing UAVs to allocate communication resources more effectively and flexibly.

The results in Figure 10 illustrate how the sum rate performance is evaluated across different UAV computational capacities for the different placement strategies. The figure shows that increasing UAV computational capacity improves the sum rate performance among all four plots, demonstrating that enhanced processing capabilities enable faster handling of computational tasks, which, in turn, optimizes communication resources, allowing for additional data transmission tasks. The SCA-based solution approach continues to yield the highest sum rate, achieving approximately 7.5 Mbps at 250 MHz. TD3-based placement achieved highly optimized sum rate of 6.7 Mbps, and the other two baseline approaches achieved a

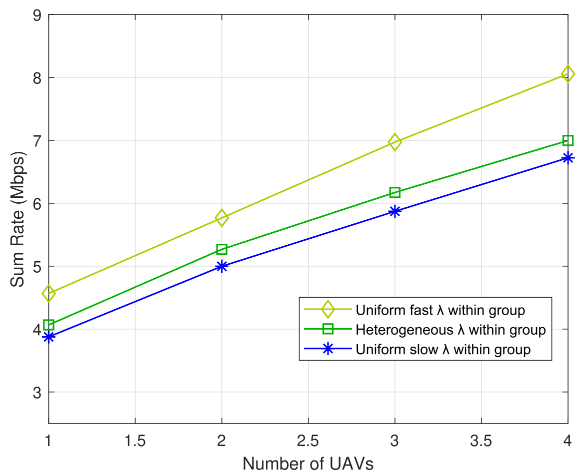

Fig. 11. Heterogeneous vs. uniform task arrival rate within a group, with $\mathcal{I} = 10$, $\mathcal{J} = 3$.

sum rate lower than 4.5 Mbps for the same computational capacity of 250 MHz.

To provide better understanding of the AoDT expression, we explore the implications of heterogeneity in the task arrival rates of devices. We consider a network with two groups of devices, where each group is composed of five devices. Fig. 11 compares different scenarios while varying the number of deployed UAVs: one scenario, referred to as *Uniform fast* λ, has a uniform and high arrival rate within each group ($\lambda_1 = 2,\ \lambda_2 = 3$), another scenario, referred to as *Uniform slow* λ, has a uniform and low arrival rate within each group ($\lambda_1 = 0.8,\ \lambda_2 = 1$), and a third scenario, referred to as *Heterogeneous* λ, has a varying arriving rate within each group ($0.8 \leq \lambda_1, \lambda_2 \leq 3$). The generated results show that the performance when considering heterogeneous task arrival rates within the same group (*Heterogeneous* λ) lies under that of *Uniform fast* λ and above that of *Uniform slow* λ. In the case of heterogeneous arrival rates, the age of the process requires status updates from all IoT devices to be aggregated thus the age will be limited by the IoT device with the slowest rate.

In summary, across all evaluated metrics (Figures 6-10), the TD3-based reinforcement learning solution consistently trails the SCA-based solution by a small margin providing clear performance gains as compared to baseline approaches. The TD3-based solution demonstrates that its learned RL policy can generalize without re-optimization, and therefore offers a scalable, operationally efficient alternative to SCA-based solution and is well-suited for larger and more dynamic networks.

VIII. Conclusion

DT technology serves as a vital tool in real-time network monitoring and performance optimization [41]. In this work, we addressed the problem of developing a highly synchronized and highly accurate digital twin of a physical system by leveraging multiple UAVs positioned in optimized locations to collect and process data from a field of IoT devices used to monitor multiple processes in the given system. Our problem considered the practical case where a group of multiple IoT devices monitors the same physical process. We formulated the problem as a mixed-integer non-convex program to optimize the total sum rate of IoT devices, resource allocation, and UAV postioning, while ensuring a constrained AoDT that reflects the digital twin data freshness. We solve the problem using successive convex approximations and propose a more efficient and scalable learning-based solution. To do so, we model the problem as MDP and solve it using twin delayed deep deterministic policy gradient method. We conduct extensive simulations to demonstrate the performance superiority of the TD3-based approach as compared to baseline solutions while remaining close to SCA results. Some interesting future research includes adopting the AoDT metric in different industry verticals including manufacturing, healthcare, aerospace, maritime and smart city environments [35]. Another interesting direction involves protecting the integrity of the digital twin against malicious attacks through securing the IoT infrastructure with block-chain-based authentication [42], [43], [44].

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